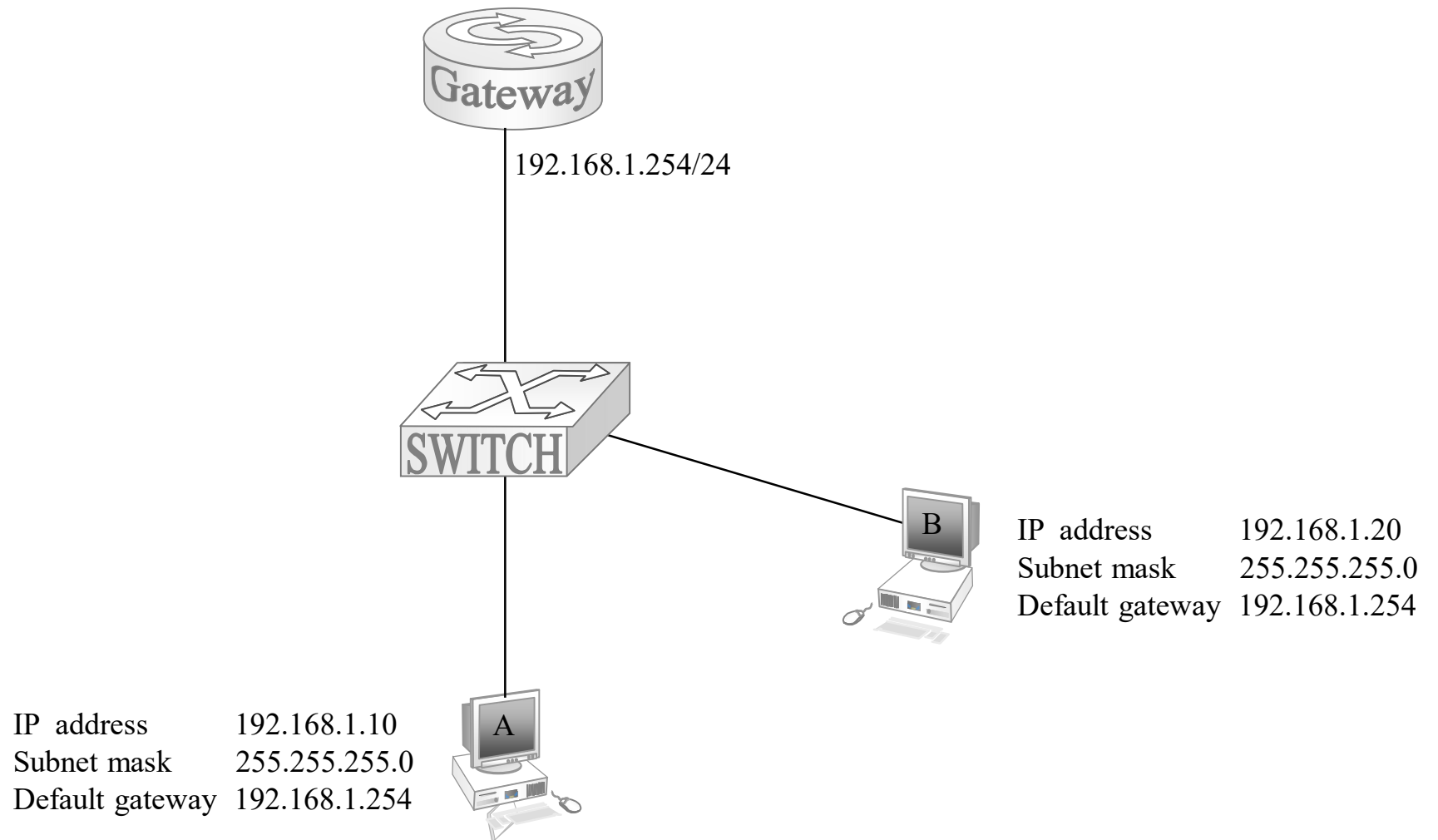
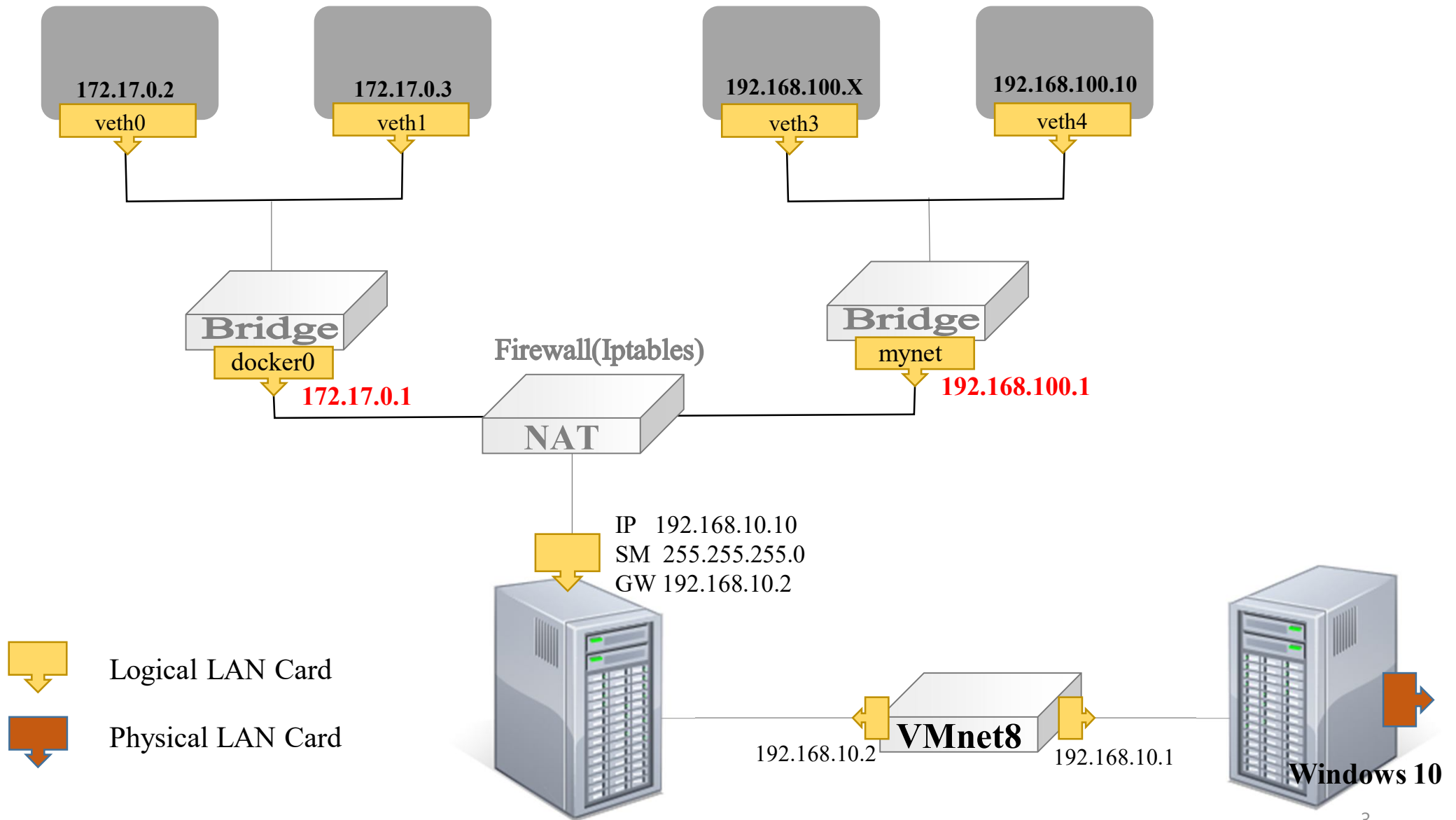


1. 컨테이너 네트워크

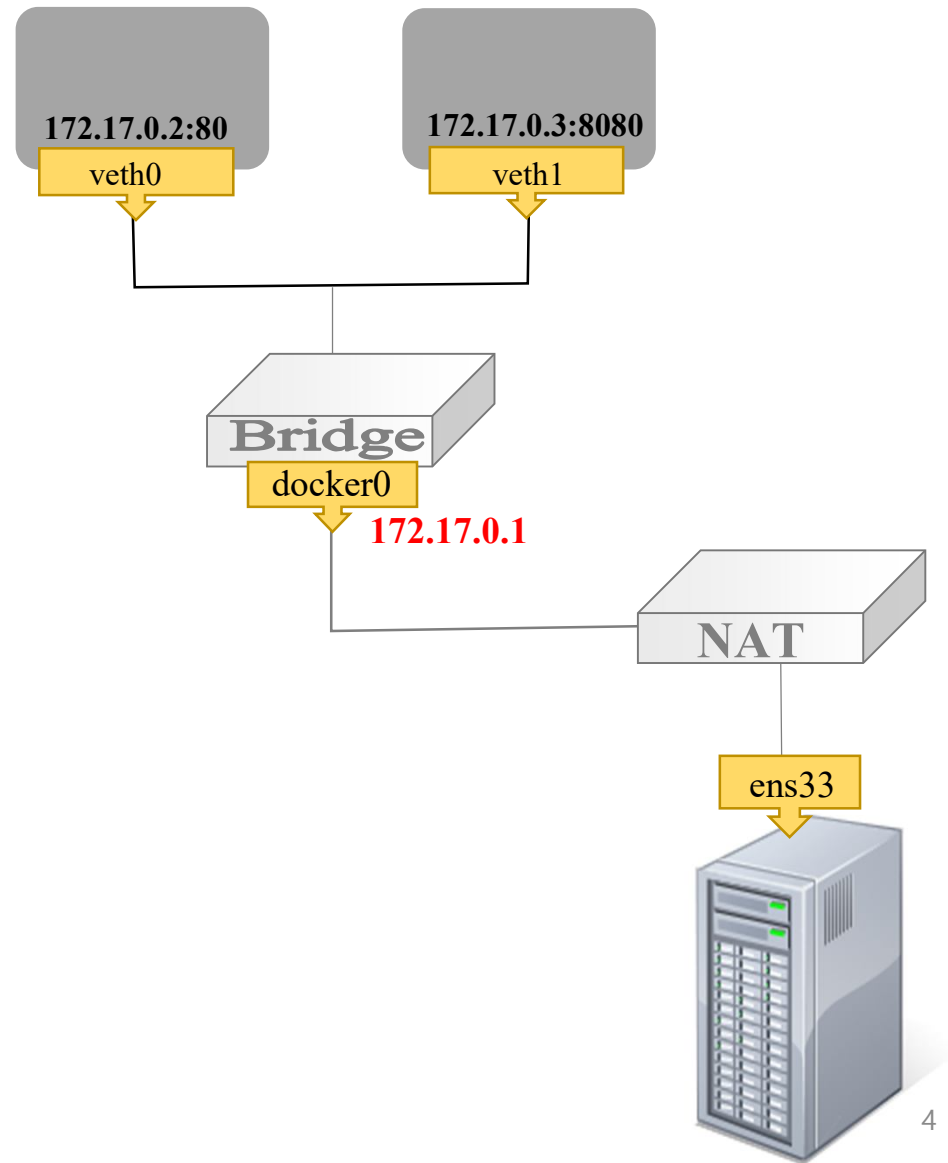
Basic Network Structure





docker0

- 모든 컨테이너들은 통신을 위해 docker0를 사용
 - Virtual Ethernet bridge : 172.17.0.0/16
- 컨테이너 생성 시 veth 인터페이스 생성(sandbox)
 - 생성된 172.16.X.X의 주소가 할당됨



ifconfig

```
docker@docker:~$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    inet6 fe80::42:51ff:fe98:11be prefixlen 64 scopeid 0x20<link>
    ether 02:42:51:98:11:be txqueuelen 0 (Ethernet)
    RX packets 177 bytes 17232 (17.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 148 bytes 17058 (17.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.10.10 netmask 255.255.255.0 broadcast 192.168.10.255
    inet6 fe80::7c5f:31d5:cfd8:713d prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:15:ce:b3 txqueuelen 1000 (Ethernet)
    RX packets 37477 bytes 44771064 (44.7 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 7069 bytes 741380 (741.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

`docker run -it --name c1 busybox`

`ip addr`

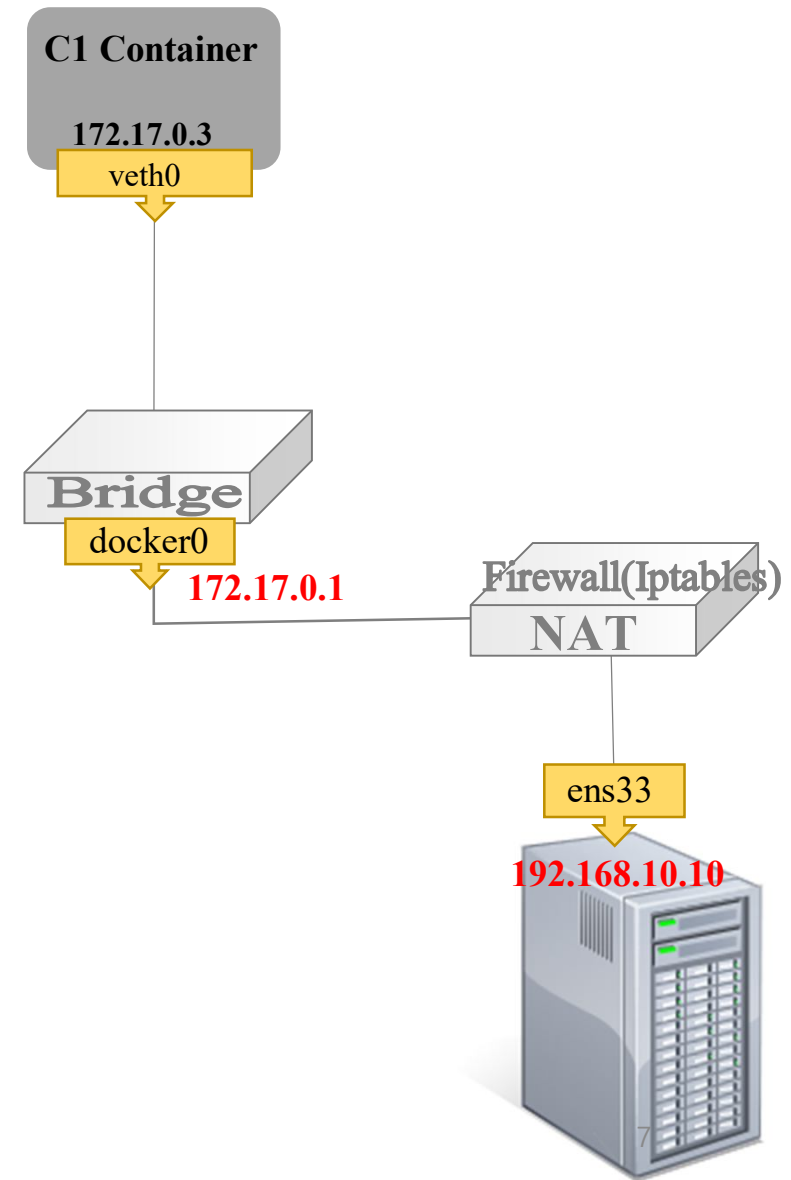
```
docker@docker:~$ docker run -it --name c1 busybox
Unable to find image 'busybox:latest' locally
latest: Pulling from library/busybox
ec562eabd705: Already exists
Digest: sha256:5eef5ed34e1e1ff0a4ae850395cbf665c4de6b4b83a32a0bc7bcb998e24e7bbb
Status: Downloaded newer image for busybox:latest
/ # ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
6: eth0@if7: <BROADCAST,MULTICAST,UP,LOWER_UP,M-DOWN> mtu 1500 qdisc noqueue
    link/ether 02:42:ac:11:00:03 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.3/16 brd 172.17.255.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::42:acff:fe11:3/64 scope link
        valid_lft forever preferred_lft forever
/ #
```

ifconfig

```
root@docker:~# ifconfig
docker0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    inet6 fe80::42:51ff:fe98:11be prefixlen 64 scopeid 0x20<link>
    ether 02:42:51:98:11:be txqueuelen 0 (Ethernet)
    RX packets 185 bytes 17776 (17.7 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 150 bytes 17318 (17.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.10.10 netmask 255.255.255.0 broadcast 192.168.10.255
    inet6 fe80::7c5f:31d5:cfd8:713d prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:15:ce:b3 txqueuelen 1000 (Ethernet)
    RX packets 37679 bytes 44787067 (44.7 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 7196 bytes 754925 (754.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

veth0de666c: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::7c6d:deff:feac:7218 prefixlen 64 scopeid 0x20<link>
    ether 7e:6d:de:ac:72:18 txqueuelen 0 (Ethernet)
    RX packets 8 bytes 656 (656.0 B)
```



docker ps

```
docker@docker:~$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
92bcffa22bcf	nginx	"/docker-entrypoint..."	7 minutes ago	Up 7 minutes	0.0.0.0:80->80/tcp, :::80->80/tcp	web1
79504f71782c	busybox	"sh"	11 minutes ago	Up 11 minutes		c2
a41194aee85b	busybox	"sh"	14 minutes ago	Up 14 minutes		c1

```
docker@docker:~$  
docker@docker:~$ docker inspect c1
```

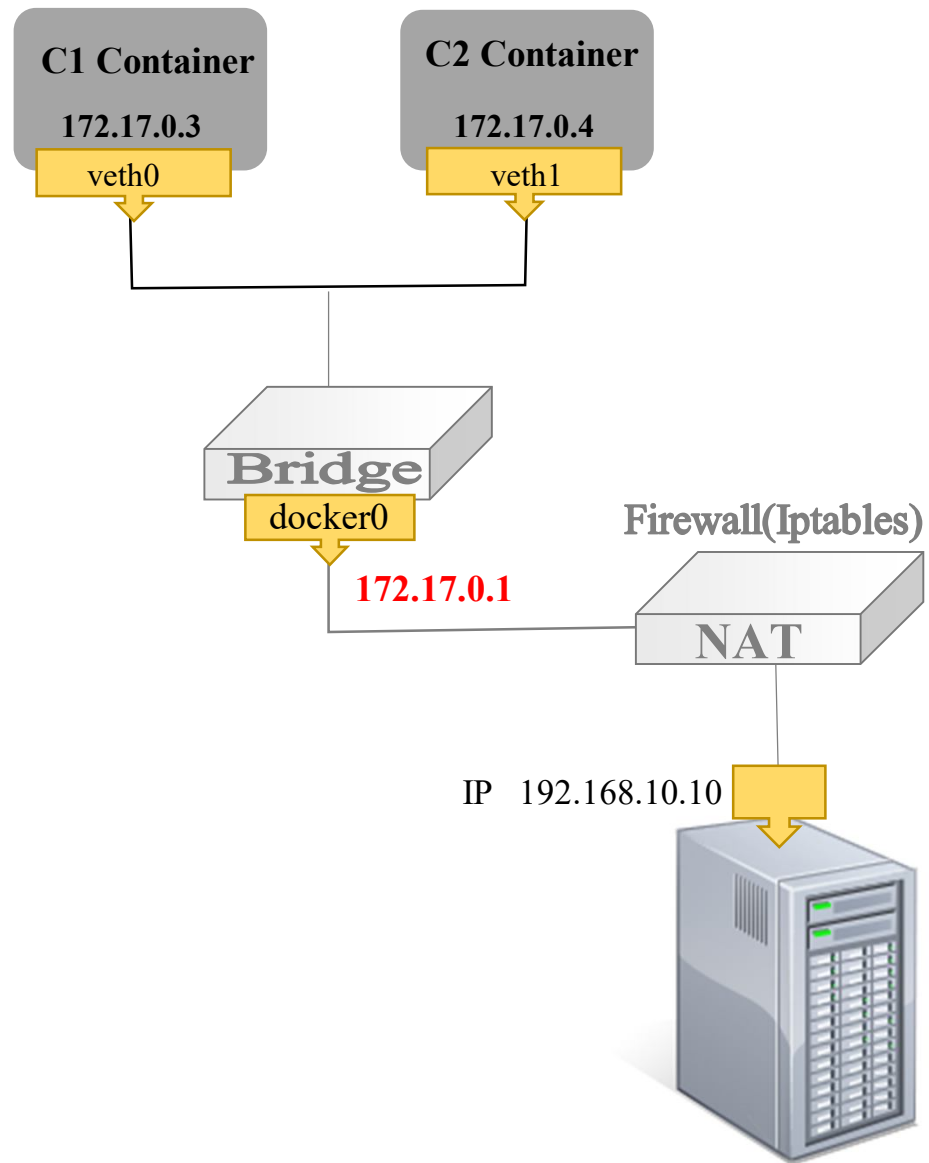
docker inspect c1

```
"NetworkSettings": {  
  "Bridge": "",  
  "SandboxID": "1fc56df88c448098e1e7bb3c7a9a529b5472e19920d2fffc63f8db44d0a99f92",  
  "SandboxKey": "/var/run/docker/netns/1fc56df88c44",  
  "Ports": {},  
  "HairpinMode": false,  
  "LinkLocalIPv6Address": "",  
  "LinkLocalIPv6PrefixLen": 0,  
  "SecondaryIPAddresses": null,  
  "SecondaryIPv6Addresses": null,  
  "EndpointID": "3438c93b8d4779798f4f878cc97c9d7cbff5873f675ad4ef38c62adabe982b9f",  
  "Gateway": "172.17.0.1",  
  "GlobalIPv6Address": "",  
  "GlobalIPv6PrefixLen": 0,  
  "IPAddress": "172.17.0.3",  
  "IPPrefixLen": 16,  
  "IPv6Gateway": "",  
  "MacAddress": "02:42:ac:11:00:03",  
  "Networks": {  
    "bridge": {  
      "IPAMConfig": null,  
      "Links": null,  
      "MacAddress": "02:42:ac:11:00:03",  
      "NetworkID": "172170002",  
      "Parent": null,  
      "PortMapping": null,  
      "Scope": "local"
```

`docker run -it --name c2 busybox`

`ip addr`

```
docker@docker:~$ docker run --name c2 -it busybox
/ # ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
8: eth0@if9: <BROADCAST,MULTICAST,UP,LOWER_UP,M-DOWN> mtu 1500 qdisc noqueue
    link/ether 02:42:ac:11:00:04 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.4/16 brd 172.17.255.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::42:acff:fe11:4/64 scope link
        valid_lft forever preferred_lft forever
/ #
```

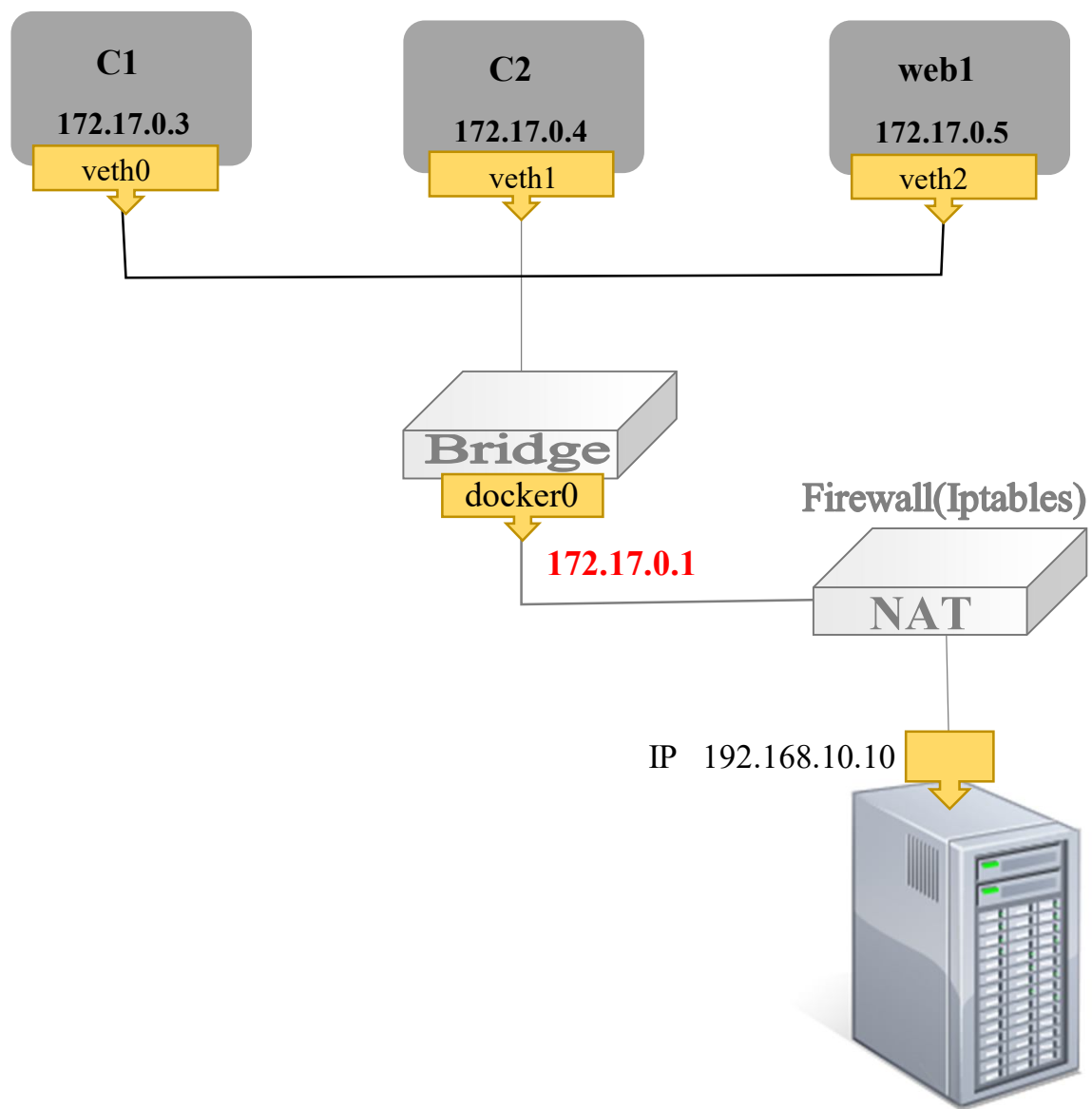


`docker run -d -p 80:80 --name web1 nginx`

```
docker@docker:~$ docker run -d -p 80:80 --name web1 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
09f376ebb190: Already exists
5529e0792248: Pull complete
9b3addd3eb3d: Pull complete
57910a8c4316: Pull complete
7b5f78f21449: Pull complete
b7923aa4e8a6: Pull complete
785625911f12: Pull complete
Digest: sha256:0f04e4f646a3f14bf31d8bc8d885b6c951fdcf42589d06845f64d18aec6a3c4d
Status: Downloaded newer image for nginx:latest
92bcffa22bcfe535c4da79f20fc303ff0b65b13d15d6879af6408afc3d2bc9cb
docker@docker:~$
```

`curl 172.17.0.5`

```
docker@docker:~$ curl 172.17.0.5
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
```



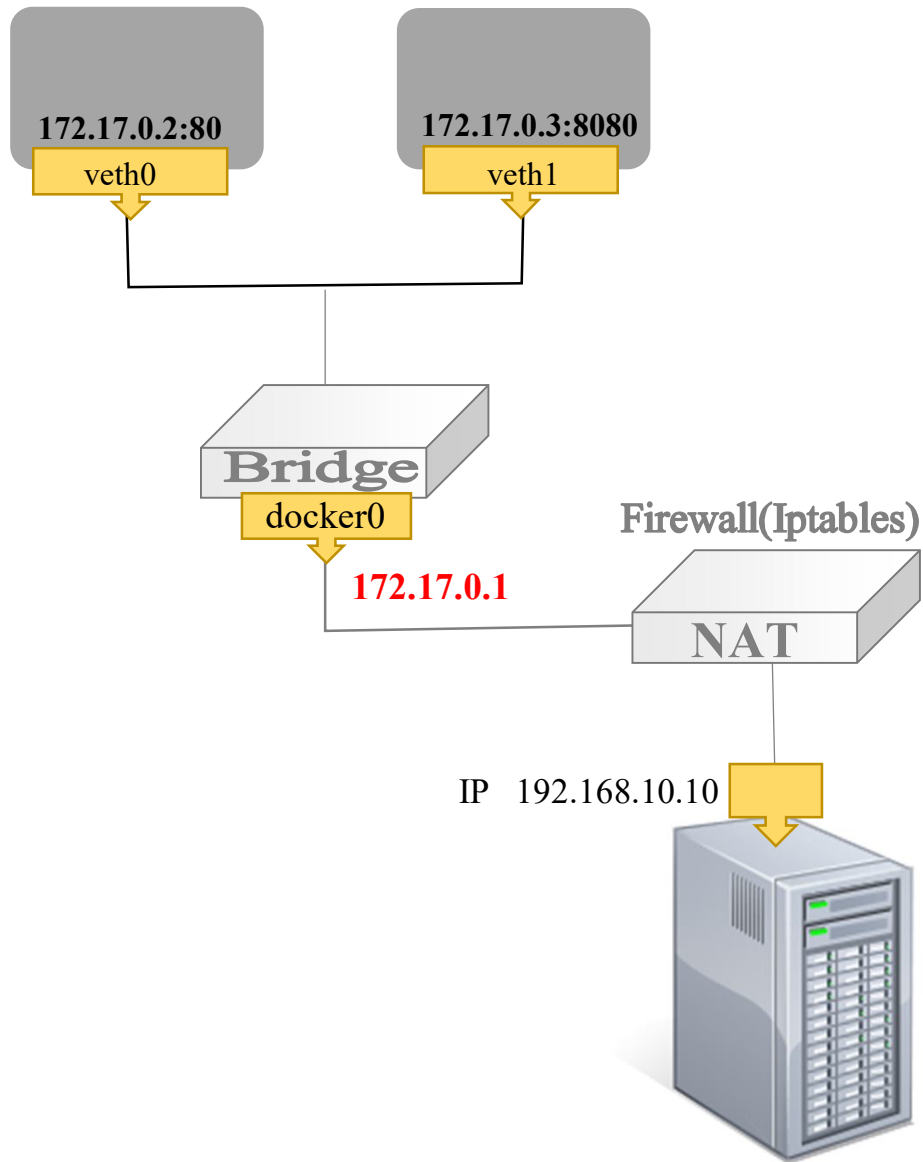
`docker rm -f $(docker ps -aq)`

`docker ps`

`docker ps -a`

```
root@docker:~# docker rm -f $(docker ps -aq)
92bcffa22bcf
79504f71782c
a41194aee85b
a33985f8f76c
root@docker:~# docker ps
CONTAINER ID    IMAGE    COMMAND    CREATED    STATUS    PORTS    NAMES
root@docker:~# docker ps -a
CONTAINER ID    IMAGE    COMMAND    CREATED    STATUS    PORTS    NAMES
root@docker:~#
```

2. 포트 포워딩(port-forwarding)



Port-forwarding

- 컨테이너 포트와 외부 호스트 포트를 연결하여 외부와 연결
- 리눅스 방화벽 iptables을 이용 → NAT 기능을 사용

- ❶ -p 호스트포트번호:컨테이너포트번호
- ❷ -p 컨테이너포트번호
- ❸ -P

(ex) `docker run --name web -d -p 80:80 nginx`

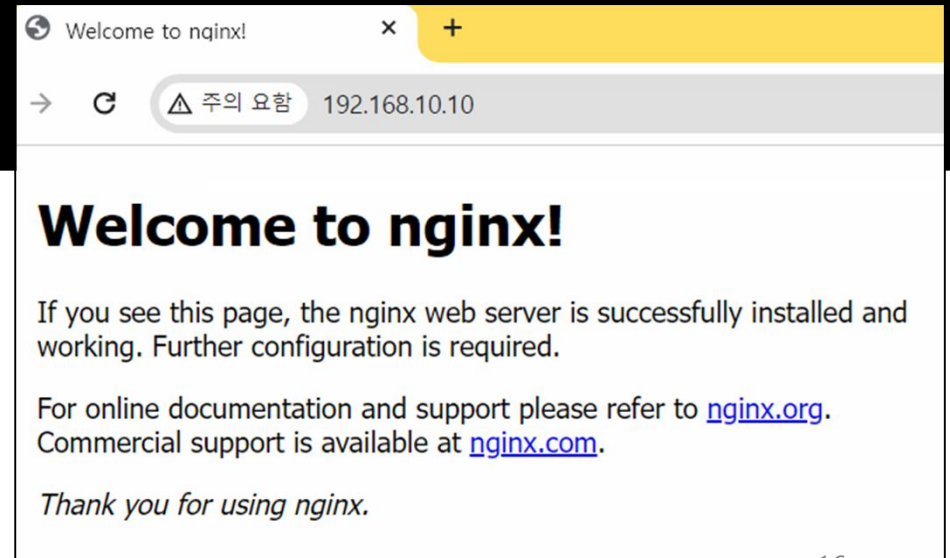
`iptables -t nat -L -n -v`

`docker run -d -p 80:80 --name web1 nginx`

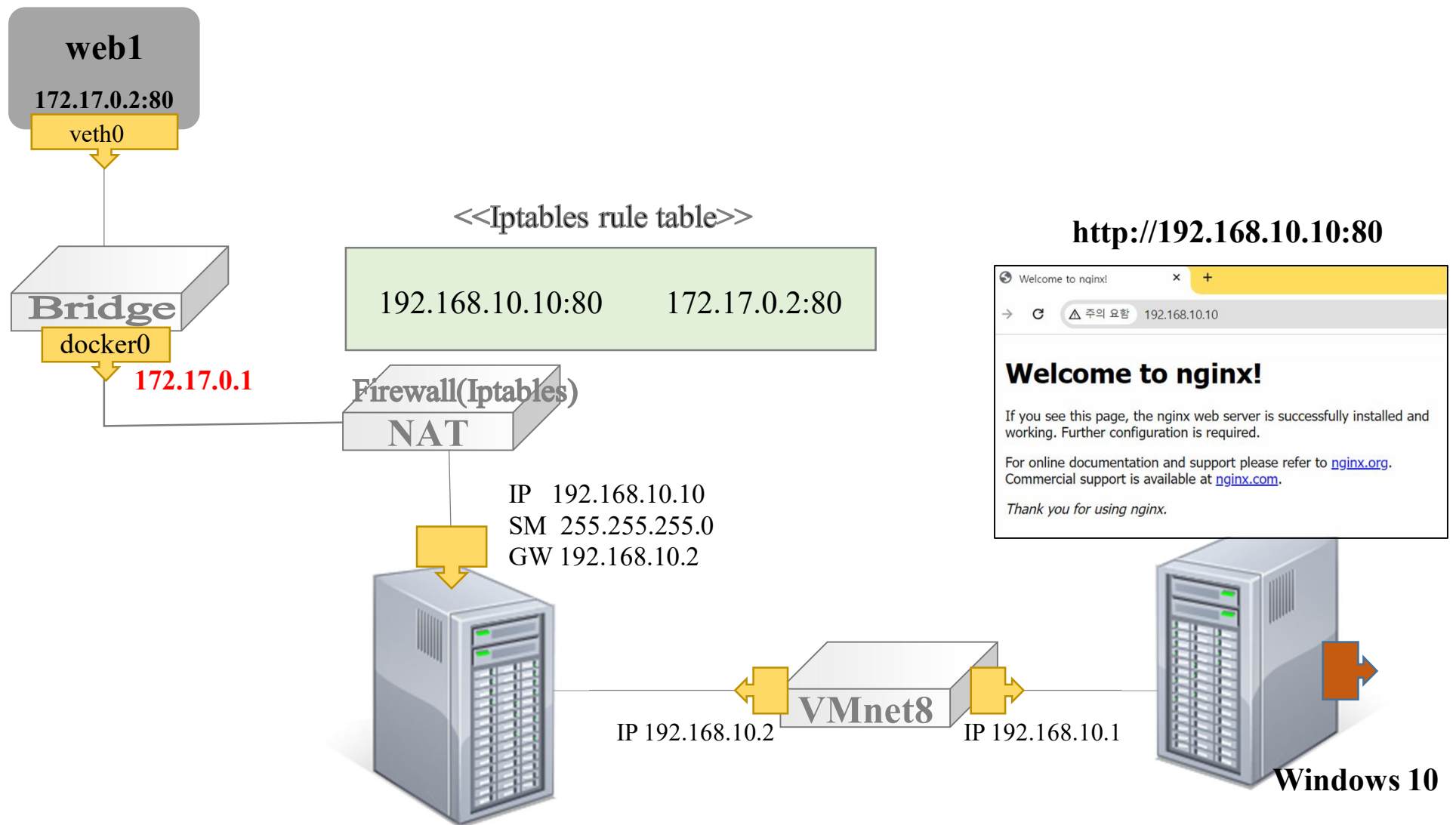
`-p` 호스트포트번호:컨테이너포트번호

`curl localhost:80`

```
docker@docker:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
09ff28f1e49d   nginx    "/docker-entrypoint..." 20 seconds ago Up 18 seconds  0.0.0.0:80->80/tcp, :::80->80/tcp  web1
docker@docker:~$
docker@docker:~$ curl localhost:80
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
```



`http://192.168.10.10:80`



sudo su -

iptables -t nat -L -v

```
docker@docker:~$ iptables -t nat -L -v
iptables v1.8.7 (nf_tables): Could not fetch rule set generation id: Permission denied (you must be root)

docker@docker:~$ sudo su -
[sudo] password for docker:
root@docker:~# iptables -t nat -L -v
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target     prot opt in     out     source    destination
    5  1465 DOCKER      all  --  any    any     anywhere  anywhere           ADDRTYPE match dst-type LOCAL

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target     prot opt in     out     source    destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target     prot opt in     out     source    destination
    0     0 DOCKER      all  --  any    any     anywhere  !localhost/8       ADDRTYPE match dst-type LOCAL

Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target     prot opt in     out     source    destination
    0     0 MASQUERADE all  --  any    !docker0 172.17.0.0/16  anywhere
    0     0 MASQUERADE tcp  --  any    any      172.17.0.5    172.17.0.5       tcp dpt:http
```

```
docker run -d -p 80 --name web2 nginx
```

-p 컨테이너포트번호

```
docker ps
```

```
curl localhost:32768
```

```
docker@docker:~$ docker run -p 80 -d --name web2 nginx
9ff6bc118b5d904fff6bbb89a1f9b65a3ff68d1881d72f4663e8b0d1d062626e
```

```
docker@docker:~$
```

```
docker@docker:~$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
9ff6bc118b5d	nginx	"/docker-entrypoint..."	47 seconds ago	Up 46 seconds	0.0.0.0:32768->80/tcp, :::32768->80/tcp	web2
09ff28f1e49d	nginx	"/docker-entrypoint..."	6 minutes ago	Up 6 minutes	0.0.0.0:80->80/tcp, :::80->80/tcp	web1

```
docker@docker:~$
```

```
docker@docker:~$ curl localhost:32768
```

```
<!DOCTYPE html>
```

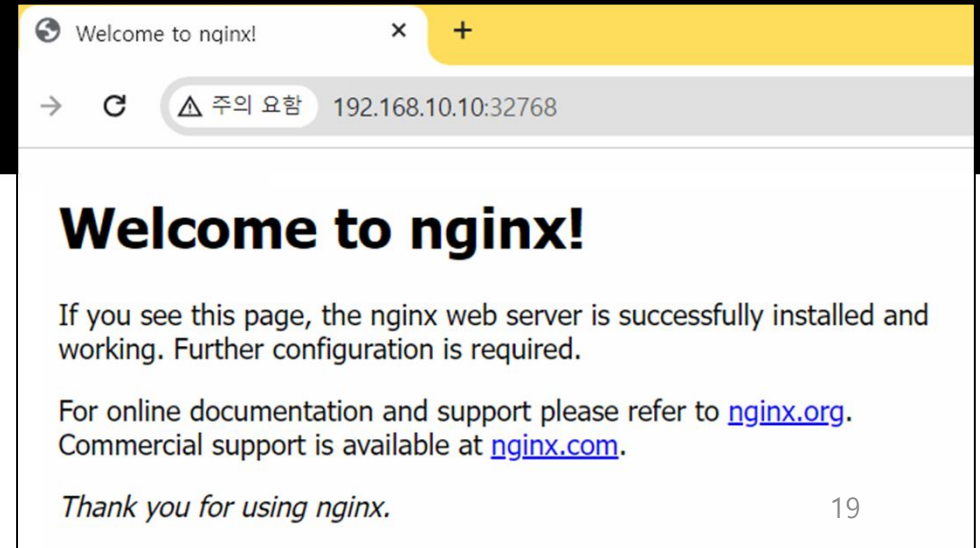
```
<html>
```

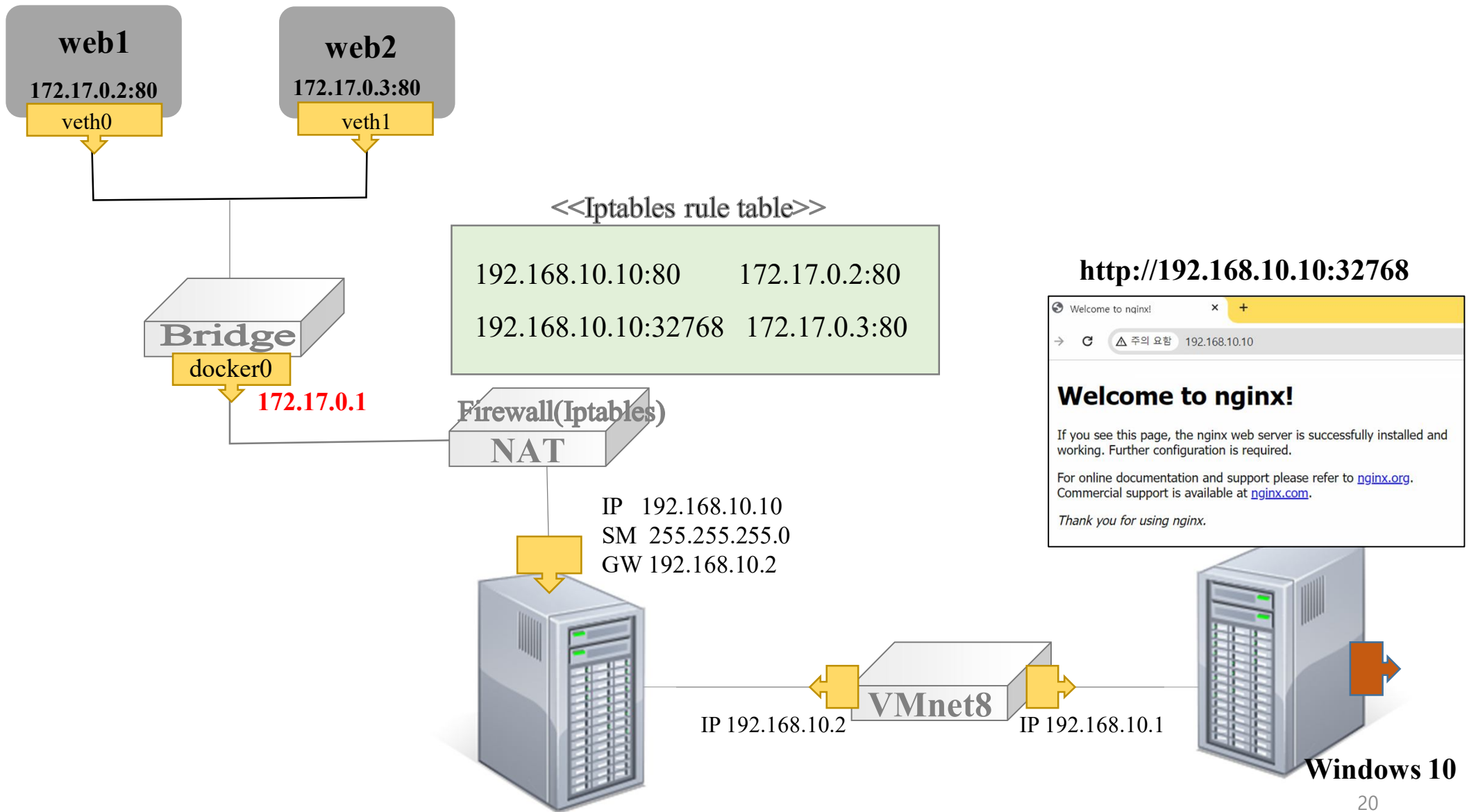
```
<head>
```

```
<title>Welcome to nginx!</title>
```

```
<style>
```

http://192.168.10.10:32768





docker run -d -P --name web3 nginx

-P

docker ps

```
docker@docker:~$ docker run -P -d --name web3 nginx
44c3adceeb781f75e37afcc50ddb822ce30fccf5506413d5ee13a2ed1247bcc0
```

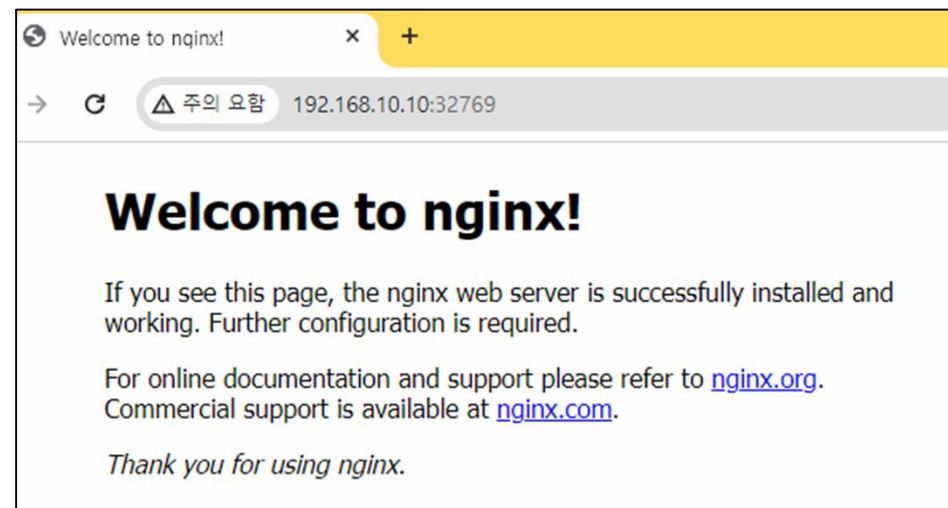
```
docker@docker:~$
```

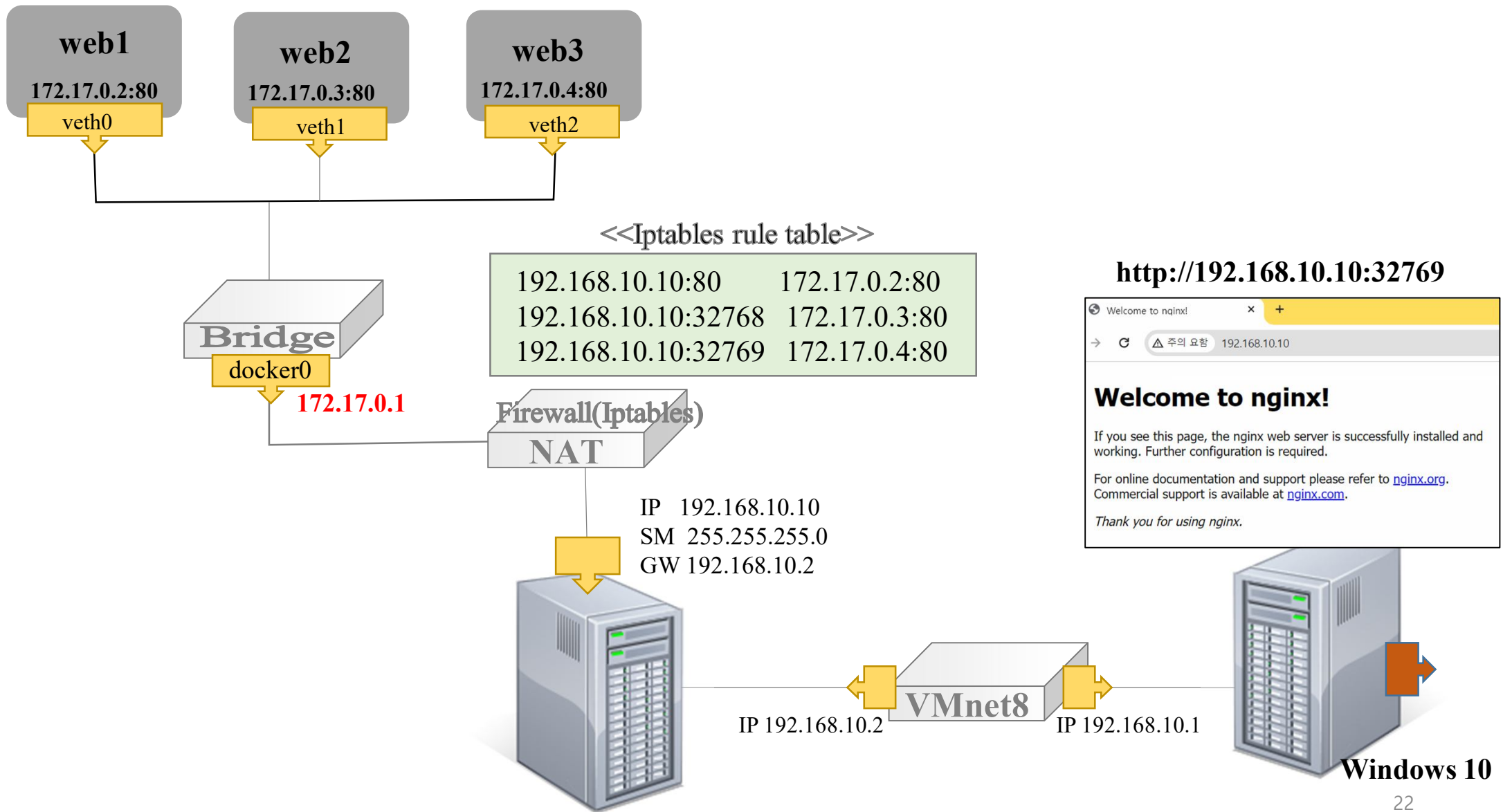
```
docker@docker:~$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
44c3adceeb78	nginx	"/docker-entrypoint..."	4 seconds ago	Up 3 seconds	0.0.0.0:32769->80/tcp, :::32769->80/tcp	web3
9ff6bc118b5d	nginx	"/docker-entrypoint..."	7 minutes ago	Up 7 minutes	0.0.0.0:32768->80/tcp, :::32768->80/tcp	web2
09ff28f1e49d	nginx	"/docker-entrypoint..."	13 minutes ago	Up 13 minutes	0.0.0.0:80->80/tcp, :::80->80/tcp	web1

```
docker@docker:~$
```

```
109 COPY docker-entrypoint.sh /
110 COPY 10-listen-on-ipv6-by-default.sh /docker-entrypoint.d
111 COPY 15-local-resolvers.envsh /docker-entrypoint.d
112 COPY 20-envsubst-on-templates.sh /docker-entrypoint.d
113 COPY 30-tune-worker-processes.sh /docker-entrypoint.d
114 ENTRYPOINT ["/docker-entrypoint.sh"]
115
116 EXPOSE 80
117
118 STOPSIGNAL SIGQUIT
119
120 CMD ["nginx", "-g", "daemon off;"]
```





```
docker rm -f $(docker ps -aq)
```

```
docker ps
```

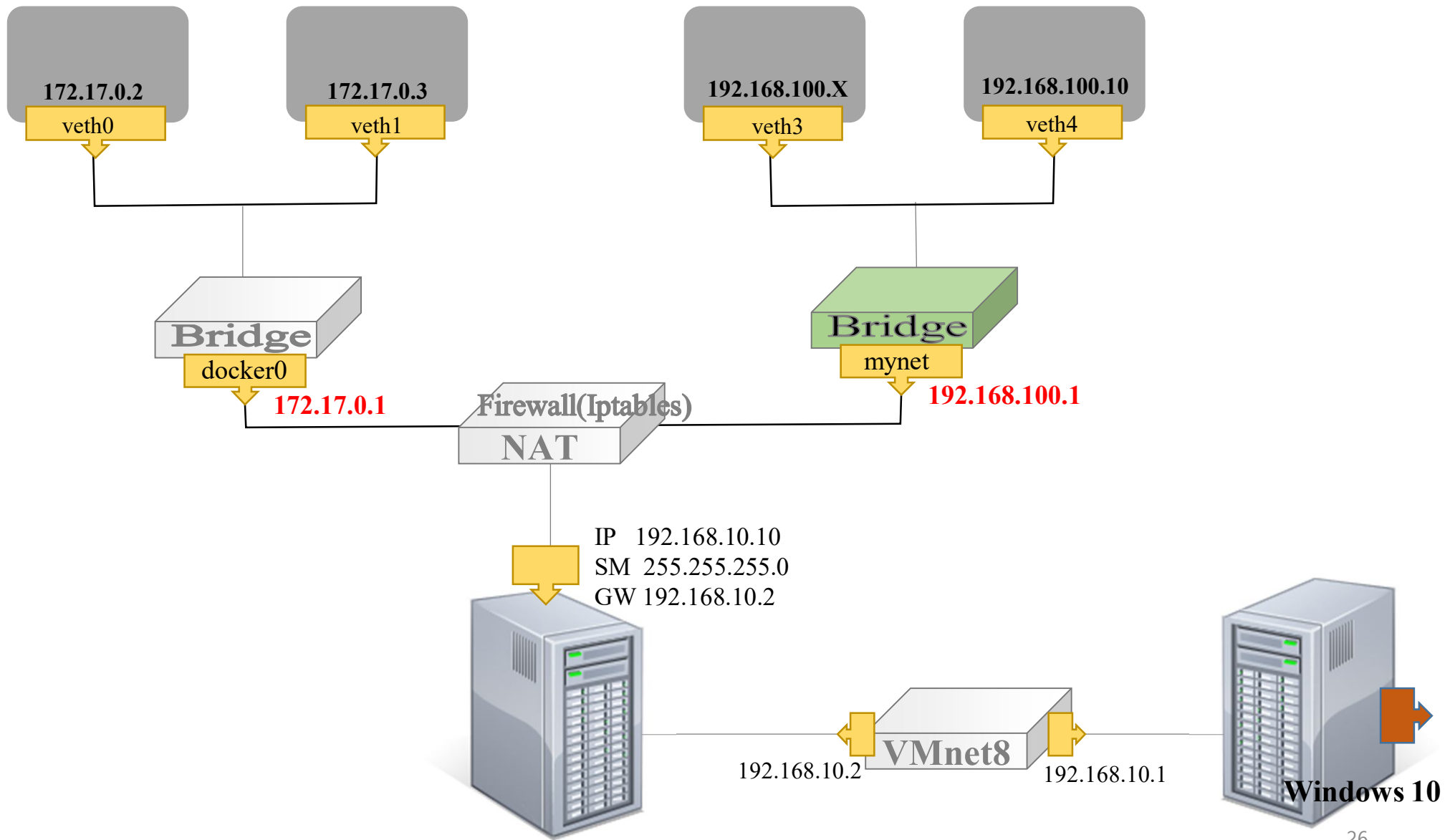
```
docker ps -a
```

```
docker@docker:~$ docker rm -f $(docker ps -aq)
docker@docker:~$ docker ps
CONTAINER ID    IMAGE    COMMAND    CREATED    STATUS    PORTS    NAMES
docker@docker:~$ docker ps -a
CONTAINER ID    IMAGE    COMMAND    CREATED    STATUS    PORTS    NAMES
docker@docker:~$
```

3. 컨테이너 네트워크 생성

Docker network 생성

- `docker network create [네트워크이름]`
- `docker network create --driver bridge [네트워크이름]`



docker network ls

docker network create --driver bridge --subnet 192.168.100.0/24 --gateway 192.168.100.1 mynet

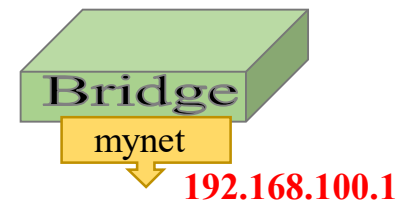
docker network ls

```
docker@docker:~$ docker network ls
NETWORK ID      NAME      DRIVER      SCOPE
bf8373bcf8d6    bridge    bridge      local
a261147a5aca    host      host        local
86231867038b    none      null        local
docker@docker:~$
docker@docker:~$ docker network create --driver bridge --subnet 192.168.100.0/24 --gateway 192.168.100.1 mynet
93144accc3b768076ad5a5e9e2de8bce4300ed2605b11e1815b517a058ce2b96
docker@docker:~$
docker@docker:~$ docker network ls
NETWORK ID      NAME      DRIVER      SCOPE
bf8373bcf8d6    bridge    bridge      local
a261147a5aca    host      host        local
93144accc3b7     mynet     bridge      local
86231867038b    none      null        local
docker@docker:~$
```

docker network inspect mynet

```
docker@docker:~$ docker network inspect mynet
```

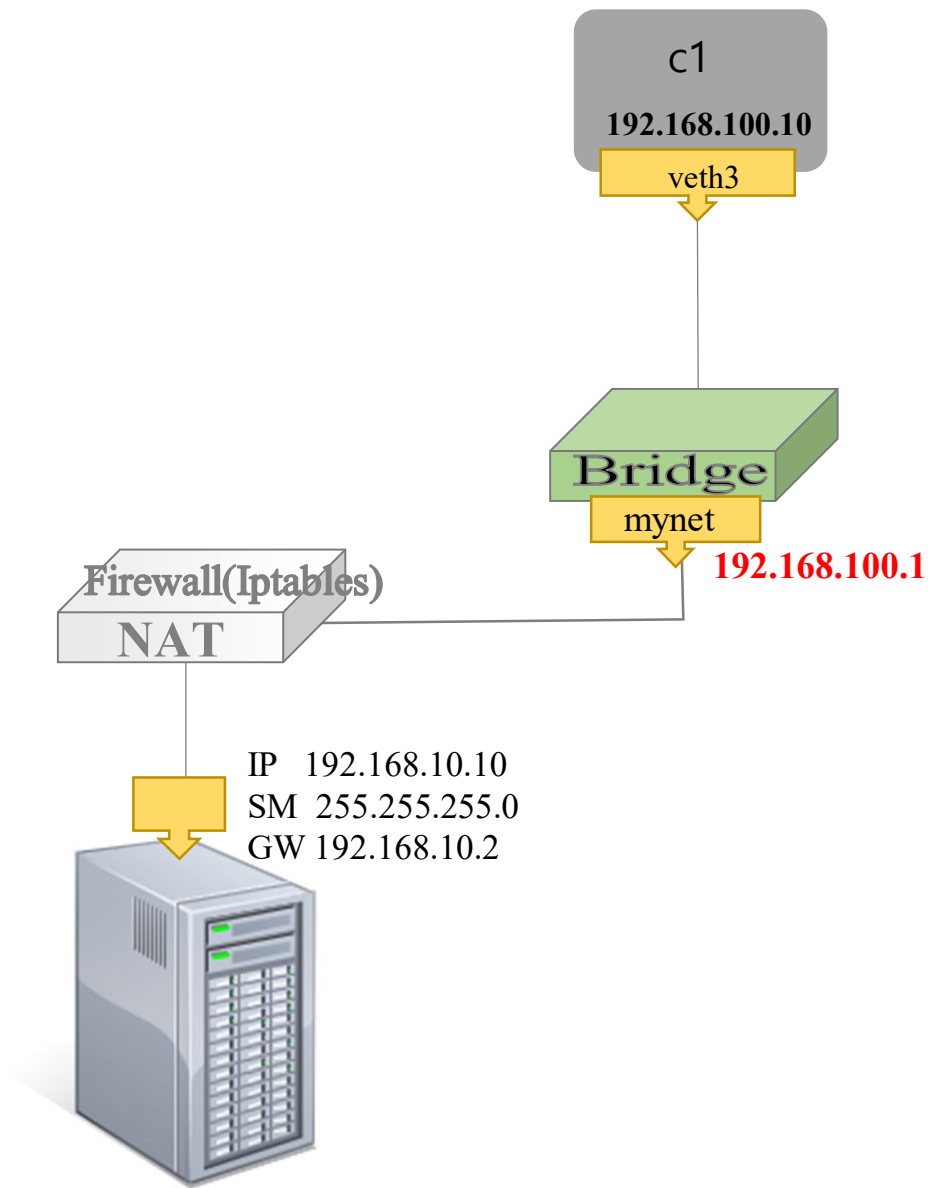
```
"EnableIPv6": false,  
"IPAM": {  
  "Driver": "default",  
  "Options": {},  
  "Config": [  
    {  
      "Subnet": "192.168.100.0/24",  
      "Gateway": "192.168.100.1"  
    }  
  ]  
}
```



`docker run -it --name c1 --net mynet --ip 192.168.100.10 busybox`

`ip addr`

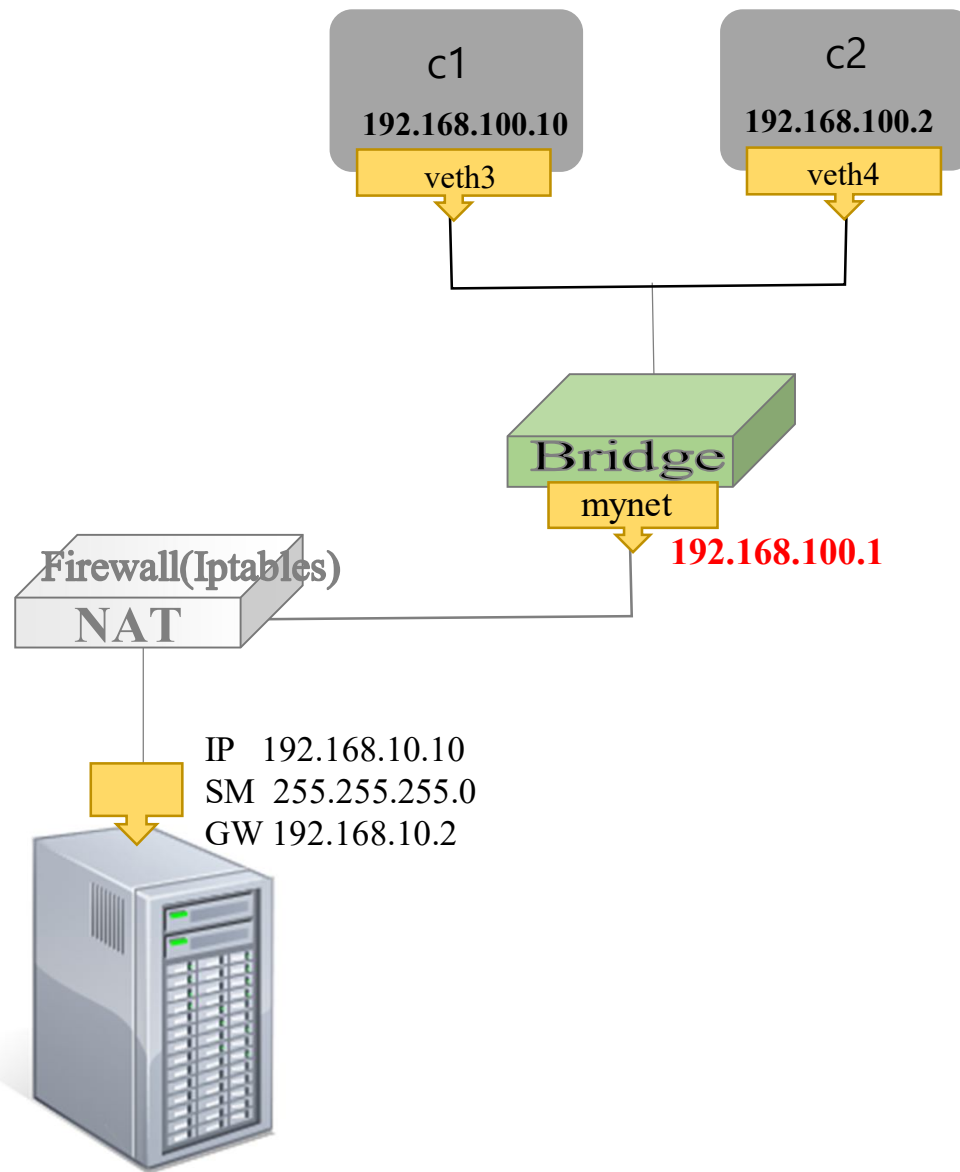
```
docker@docker:~$ docker run -it --name c1 --net mynet --ip 192.168.100.10 busybox
/ # ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
19: eth0@if20: <BROADCAST,MULTICAST,UP,LOWER_UP,M-DOWN> mtu 1500 qdisc noqueue
    link/ether 02:42:c0:a8:64:0a brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.10/24 brd 192.168.100.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::42:c0ff:fea8:640a/64 scope link
        valid_lft forever preferred_lft forever
/ #
```



`docker run -it --name c2 --net mynet busybox`

`ip addr`

```
docker@docker:~$ docker run -it --name c2 --net mynet busybox
/ # ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
21: eth0@if22: <BROADCAST,MULTICAST,UP,LOWER_UP,M-DOWN> mtu 1500 qdisc noqueue
    link/ether 02:42:c0:a8:64:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.2/24 brd 192.168.100.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::42:c0ff:fea8:6402/64 scope link
        valid_lft forever preferred_lft forever
/ #
```



Docker network 삭제

- `docker network rm` [네트워크이름]

Docker network 삭제

```
docker rm -f $(docker ps -aq)
```

```
docker ps
```

```
docker ps -a
```

```
docker network rm mynet
```

```
docker network ls
```